

Note 3.5 Change of Basis

Changing Coordinates in $\mathbb{R}^2$

For any vector $\mathbf{x}$ in $\mathbb{R}^2$, it can be represented as the linear combination that

$\mathbf{x} = c_1 \mathbf{e}_1 + c_2 \mathbf{e}_2$ where $\{\mathbf{e}_1, \mathbf{e}_2\}$ is the standard basis in $\mathbb{R}^2$.

The scalars c_1, c_2 are the **coordinates** of $\mathbf{x}$ with respect to the standard basis. By [Theorem 3.3.2](#), for any basis $\{\mathbf{y}, \mathbf{z}\}$ in $\mathbb{R}^2$, there is a unique linear combination

$$\mathbf{x} = \alpha \mathbf{y} + \beta \mathbf{z}$$

So we can call the scalars α, β as the coordinates of $\mathbf{x}$ with respect to the basis $\{\mathbf{y}, \mathbf{z}\}$. Also, we can call the vector $(\alpha, \beta)^T$ as the **coordinate vector** of $\mathbf{x}$ with respect to $[\mathbf{y}, \mathbf{z}]$, which is the notation of ordered basis.

Application I Population Migration

For the population with fixed immigration and emigration rate between the city and the suburbs, we can use the matrix multiplication. Suppose $\mathbf{x}_n$ is the vectors containing the population in both city and suburbs after n years, then $\mathbf{x}_n = A^n \mathbf{x}_0$. Then as n increases, the sequence converges to a limit $\mathbf{x}$, or called **steady-state vector**. We can understand the process by using a different coordinate system. We can pick vectors $\mathbf{u}_1, \mathbf{u}_2$ such that

$$\mathbf{x}_0 = c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2$$

and

$$A\mathbf{u}_1 = \mathbf{u}_1, A\mathbf{u}_2 = k\mathbf{u}_2$$

where $\mathbf{u}_1$ is any multiple of the limit vector $\mathbf{x}$.

Then

$$\mathbf{x}_n = A^n \mathbf{x}_0 = c_1 \mathbf{u}_1 + c_2 k^n \mathbf{u}_2$$

In general, the $c_2 k^n$ is negative and k is less than 1. So as n increases, $\mathbf{x}_n \rightarrow c_1 \mathbf{u}_1$.

And this model is called **Markov process** and the sequence $\mathbf{x}_1, \mathbf{x}_2, \dots$ is called a **Markov chain**.

Changing Coordinates

When changing coordinates, we need to consider the following two problems:

- How to find the coordinates of a vector $\mathbf{x} = (x_1, x_2)^T$ (or, $x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2$) with respect to $\mathbf{u}_1, \mathbf{u}_2$.
- How to find the coordinates of a vector $c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2$ with respect to $\mathbf{e}_1, \mathbf{e}_2$.

Second Problem

Suppose that

$$\mathbf{x} = c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2$$

That is

$$\mathbf{x} = U \mathbf{c}$$

where $U = (\mathbf{u}_1, \mathbf{u}_2)$ and the coordinate vector $\mathbf{c} = (c_1, c_2)^T$. Here the matrix is called **transition matrix** from the ordered basis $\{\mathbf{u}_1, \mathbf{u}_2\}$ to the basis $\{\mathbf{e}_1, \mathbf{e}_2\}$.

First Problem

For the first problem, we need to find the transition matrix from the basis $\{\mathbf{e}_1, \mathbf{e}_2\}$ to the basis $\{\mathbf{u}_1, \mathbf{u}_2\}$. Since U is nonsingular, so we have

$$\mathbf{c} = U^{-1} \mathbf{x}$$

Thus given a vector $\mathbf{x}$, we just need to multiply by U^{-1} to find the coordinate vector with respect to $\{\mathbf{x}_1, \mathbf{x}_2\}$. Here U^{-1} is the transition matrix from $\{\mathbf{e}_1, \mathbf{e}_2\}$ to $\{\mathbf{u}_1, \mathbf{u}_2\}$.

General Changing between Two basis in $\mathbb{R}^2$

Suppose we want to change a vector $\mathbf{x}$ from one basis $[\mathbf{v}_1, \mathbf{v}_2]$ to another basis $[\mathbf{u}_1, \mathbf{u}_2]$.

Suppose $\mathbf{x} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2$ and $\mathbf{x} = d_1 \mathbf{u}_1 + d_2 \mathbf{u}_2$

Then since

$$\mathbf{x} = V \mathbf{c} \quad \mathbf{d} = U^{-1} \mathbf{x}$$

Then we have

$$\mathbf{d} = U^{-1} V \mathbf{c}$$

Here the matrix $S = U^{-1} V$ is the transition matrix.

Change of Basis for a General Vector Space

Definition

Let V be a vector space and $E = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be an ordered basis for V . If $\mathbf{v}$ is any element of V , then

$$\mathbf{v} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n$$

where $c_1, c_2, \dots, c_n$ are scalars. The vector $\mathbf{c} = (c_1, c_2, \dots, c_n)^T$ defined in this way is called **coordinate vector** of $\mathbf{v}$ with respect to the ordered basis E , denoted by $[\mathbf{v}]_E$. And the c_i 's are called the **coordinates** of $\mathbf{v}$ relative to E .

And the transition matrices will become $n \times n$.

Transition Matrix

Similar to the case in two dimension, if we let $\mathbf{x} = [\mathbf{v}]_E$ and $\mathbf{y} = [\mathbf{v}]_F$ where $E = \{\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_n\}$ and $F = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$, Then we have

$$\mathbf{y} = S\mathbf{x}$$

The matrix S is the transition matrix from E to F .

And we can get the matrix S by expressing each basis vector $\mathbf{w}_j$ as a linear combination of the $\mathbf{v}_i$'s.